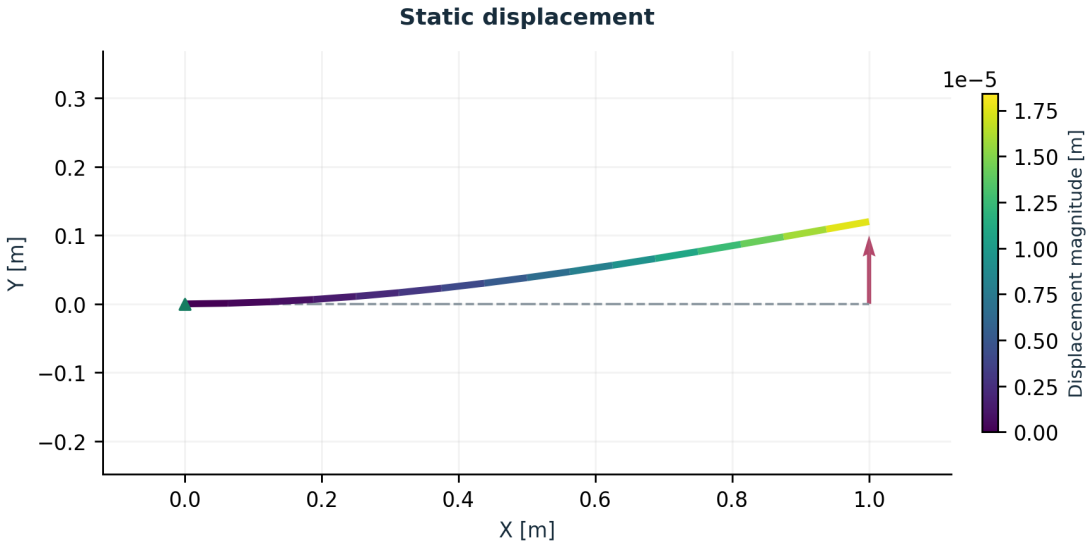
Short cantilever: Timoshenko shear flexibility

ACA02 | SOL 101 | Independent analytical equation verification

A short clamped beam separates bending and shear deflection. The effective shear area is explicitly prescribed, so the exact Timoshenko tip response is independent of mesh resolution.

Geometry / mesh	Length 1 m; section 0.04 m by 0.2 m. Meshes of 1, 2, 4, 8 and 16 CBAR elements.
Material / section	E = 70 GPa, nu = 0.3; A = 0.008 m2, I = 2.666666667e-5 m4, effective shear area 5A/6.
Boundary conditions	Every node ux=uz=rx=ry=0; root uy=rz=0. Other uy and rz free.
Applied load	+Y tip force 100 N; linear static response.



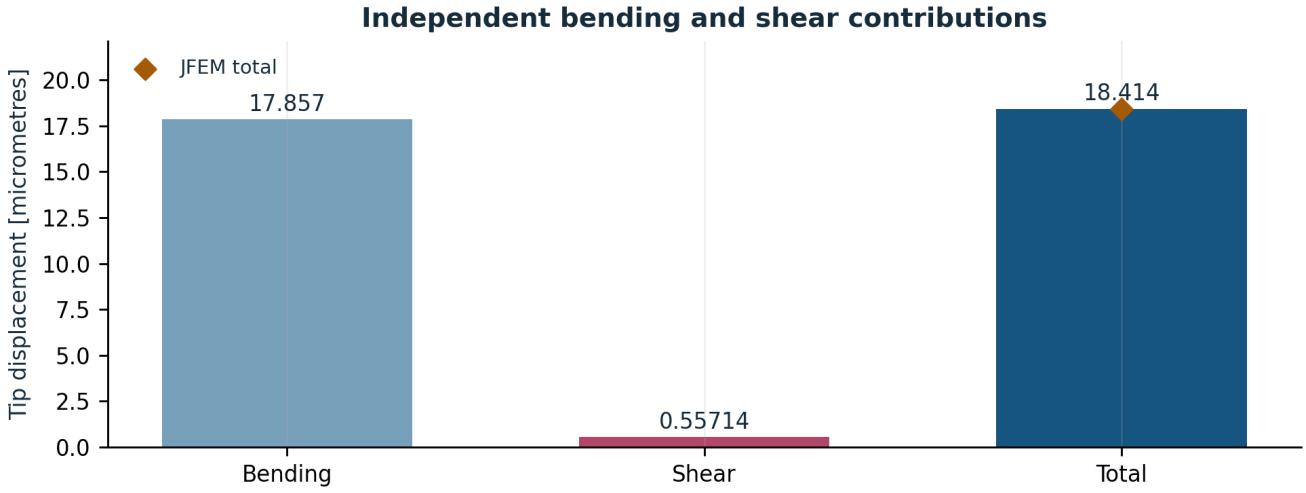
Shape amplified x6.52e+03; grey = original.

Actual solver output. Deformation is amplified for visibility; mode vectors have arbitrary sign and amplitude.

Recorded numerical checks

Quantity [unit]	JFEM	Reference	Error	Assessment
tip transverse displacement [m]	1.8414e-05	1.8414e-05	1.3413e-10%	PASS
tip section rotation [rad]	2.6786e-05	2.6786e-05	1.3877e-10%	PASS

Evidence, convergence and interpretation



Independent reference equations

G=E/(2*(1+nu))

A_s=kappa*A, kappa=5/6 explicitly specified

w_tip=F*L^3/(3*EI)+F*L/(kappa*G*A)

theta_tip=F*L^2/(2*EI)

Every mesh reproduces the exact displacement and section rotation within 1.4e-10%.

Shear contributes 3.026% of the total tip displacement; an Euler-Bernoulli model omits this contribution.

Scope: Verification of the chosen one-dimensional shear-flexible equations and prescribed effective shear area; no exact three-dimensional elasticity claim.

Reference and reproduction

University of Alberta: Timoshenko Beam
https://engcourses-uofa.ca/books/introduction-to-solid-mechanics/plane-beam-approximations/timoshenko-beam/

University of Alberta: Euler Bernoulli Beam
https://engcourses-uofa.ca/books/introduction-to-solid-mechanics/plane-beam-approximations/euler-bernoulli-beam/

Open the recorded .jfem file with Boxer: Project > Open JFEM results. For older eigenvalue files choose modal or buckling results when requested. The input deck and the recorded run receipt accompany the figures.

Run: 2026-10-19 | Solver revision: a70f997cce40 | Full precision values: results.json / metrics.csv